



Hands-on Principal Engineer · 19+ yrs · Python + Node.js + Java · Agentic AI in Production (LangGraph · MCP · Guardrails) · Governed AI-Assisted Delivery (Claude Code + OpenAI Codex) · Cloud-Native AWS · Multi-Tenant SaaS · HIPAA, ISO 27001

SUMMARY

Hands-on principal engineer, 19+ years, fluent in Python (FastAPI, LangChain, LangGraph) and Node.js, with Go and Java + Spring Boot — Java about 11 years across three eras — also in production. Shipped ten analytics and artificial intelligence (AI) services for a chronic-care platform on Amazon Web Services (AWS): a live multi-agent pipeline on LangGraph with 3 model-backed finder agents, field-level claim-grammar guardrails, and real Model Context Protocol (MCP) servers for large language model (LLM) tool use. Ran a governed Claude Code and OpenAI Codex practice across 20 engineers under hands-on review. Set Kubernetes and Terraform as the company standard, taking deploy time from hours to under 10 minutes and infrastructure cost down 25–30%. Earlier, consolidated approximately 300 enterprise instances onto multi-tenant software as a service (SaaS) with United States–India teams, and raised hotel booking conversions 65%.

Key Skills

Technology Leadership & Strategy

- Technology Strategy & Architecture
- Cross-Domain Solution Architecture (End-to-End Design Across Sub-Domains and Systems)
- Governing AI-Assisted Development Across an Engineering Organisation
- Decision-Intelligence Platforms (Event-Sourced + Graph + AI)
- Mentoring Engineers Across Levels (Design Reviews, AI Fluency)
- Cross-Functional Collaboration Across Geographies & Time Zones (Product, Quality Assurance (QA), Analytics)

Engineering Excellence

- Expert Fluency in Python (Django, FastAPI, LangChain, LangGraph) and Node.js (TypeScript)
- Agentic AI in Production (LangGraph State Graphs, Multi-Agent Finder Pipelines, Model Context Protocol (MCP) Servers for LLM Tool Use)
- LLM Evaluation & Guardrail Design (Retrieval Evaluation Harnesses, Field-Level Claim Grammar, Citation-Faithfulness Gates, Replay Determinism)
- AI Code Generation in the Delivery Workflow (Claude Code, OpenAI Codex — Governed Under Hands-On Review)
- Multi-Tenant SaaS Architecture (Shared-Schema, Per-Tenant Isolation, Regulated Carve-Outs)
- Java + Spring Boot Microservices (Java 8/17, Spring Boot 2.x, JSP — 11 years across three eras)
- Go Production Services (Node.js to Go Chat Service Port, Burst-Tested at 100,000 Connections)
- System Design at Scale (Distributed Systems + RDBMS + APIs)
- Cross-Team Technical Standards (6 sub-teams, company-wide Kubernetes, Terraform, Go and DynamoDB adoption)
- Cloud-Native AWS (Kubernetes, Terraform, Elastic Container Service (ECS) Fargate, Kinesis, Lambda, DynamoDB, Aurora, SageMaker, OpenSearch)
- Infrastructure as Code & Resilience (Multi-Availability-Zone Failover, Disaster Recovery, Load Balancing)
- Observability & Reliability Engineering (Datadog, Amplitude, Elasticsearch/Logstash/Kibana (ELK), Error Rate Under 3%)
- DevOps & Continuous Integration / Continuous Delivery (CI/CD) Transformation (Jenkins to GitHub Actions, Puppeteer Test Automation)
- Shared Platform APIs (Contracts, Versioning, Backward Compatibility)
- Event Sourcing + Command Query Responsibility Segregation (CQRS) + Snapshot Exporters
- Statistical Rigor & Leakage-Safe Machine Learning (Propensity Matching, Standardised Mean Difference Gates, Bootstrap CI, As-Of Featurisation, Outcome-Maturity Windows)
- Self-Hosted Models In-Virtual Private Cloud (VPC) (Ollama, Llama 3.1 8B/70B Cascade, SageMaker Embeddings, Self-Hosted LangSmith)
- Vector & Columnar Data Stores (Elasticsearch dense_vector, OpenSearch, DuckDB + Parquet, Redis, Amazon Neptune + Gremlin)
- AI/ML in Product (Graph Neural Network (GNN) on Amazon Neptune, LLM clinical workflows, vector retrieval)

Business & Delivery

- Agile / Scrum Delivery Management
- Commerce & Subscription Domain Delivery (Hotel Booking Funnel, 25-Supplier Integration, Subscription-Gated SaaS Entitlements)
- Security, Compliance & Identity (Health Insurance Portability and Accountability Act (HIPAA) Pipelines in an ISO 27001-Certified Environment, OpenID Connect, OAuth 2.0, Role-Based Access Control (RBAC), Least-Privilege)
- Performance Optimization & Cost Management (Infrastructure Cost Down 25–30%, Pricing Cache Cost Down 46%)
- Accessibility Compliance (WCAG 2.2 Level AA)
- Distributed United States–India Delivery Teams (Cross-Geography, Cross-Time-Zone Collaboration)

PROFESSIONAL EXPERIENCE

Principal Software Architect	Mar '25 - Apr '26
Voltuswave Technologies India Pvt Ltd	Hyderabad
• Drove a governed Claude Code and OpenAI Codex practice across a 20-engineer team — design exploration, first-cut implementation and test scaffolding, all under hands-on review	

- Orchestrated **3 model-backed finder agents** on a deterministic **LangGraph** state graph, live in production over **9–10 years** of patient history, narrator off by default
- Engineered a retrieval-augmented generation (RAG) pipeline in **Python and FastAPI** on **Elasticsearch dense_vector** search across **80+ chronic conditions**, keeping **Amazon Neptune** off the serving path
- Provisioned **AWS** on **Kubernetes** and **Terraform**, adopted as company-wide standard, with **ECS Fargate at 70% Spot** — deploy time **under 10 minutes**, infrastructure cost down **25–30%**
- Architected a two-stream **Amazon Kinesis** design with Enhanced Fan-Out, expanding **4 to 1,200 shards**, with multi-availability-zone failover for the chat platform's **HIPAA** certification
- Scaled a chat service ported from **Node.js** to **Go** to **100,000 concurrent connections** in k6-validated burst testing at **P95 under 16 ms**
- Deployed **Datadog** observability across production services, cutting the error rate to **under 3%** and making reliability measurable for the first time

Lead SDE/Sr. Staff Engineer / Staff Software Engineer

Dec '19 - Feb '25

Deque Software

Hyderabad

- Implemented multi-tenant isolation across **approximately 300 enterprise instances**, adding an organisation ID to every table and resolving tenant from the **JSON Web Token (JWT)**
- Steered **70–80%** of customers onto shared instances while regulated banks kept dedicated databases, with one codebase serving both deployment models
- Forged a central authentication service in **Java and Spring Boot**, migrating every instance from **Keycloak** to **Red Hat single sign-on** with zero user disruption
- Productized a billing and subscription service in **Node.js/TypeScript**, gating feature access by tier for **every product**, later ported to **Java and Spring Boot**
- Stewarded a **zero-downtime** upgrade from **PostgreSQL 9.6** to **Amazon Aurora PostgreSQL 13.x** across every axe Auditor instance
- Championed decomposition of a monolith into **10–15 services**, splitting on independent scaling and database ownership rather than on team boundaries
- Modernised **CI/CD** from **Jenkins** to **GitHub Actions**, achieving **ISO 27001** and **50% faster** release cycles
- Slashed regression testing time **over 50%** with an end-to-end automated framework built on **Puppeteer** in **Node.js**

Software Engineer III

Aug '18 - Jul '19

Rocket Software

Bangalore

- Anchored decomposition of **IBM Db2 Configuration Manager** into **Java and Spring Boot** microservices, defining boundaries across **z/OS and Linux, UNIX and Windows** platforms
- Coached a **5-engineer** team through delivery planning, code review and performance appraisals while remaining hands-on as an individual contributor
- Surfaced defect hot-spots through **REST** interfaces and service isolation, reducing reported bugs **over 50%**
- Partnered with product to extend platform support to **Windows** and Linux — Ubuntu, CentOS and **Red Hat Enterprise Linux (RHEL)** — raising sales **over 20%**

Co-Founder and VP of Technology

Mar '17 - Apr '18

Voltuswave Technologies India Pvt Ltd

Hyderabad

- Co-founded VoltusWave and launched its first **application Platform-as-a-Service (aPaaS)** product for web and Android, winning **8 early enterprise clients**
- Mobilized an engineering organisation from **0 to 35** in 13 months, mentoring freshers and senior engineers while establishing delivery process
- Helmed development, deployment and operations on **Amazon Elastic Container Service**, setting the architecture roadmap every subsequent product inherited

Senior Software Engineer

Mar '16 - Mar '17

Teletext India Private Limited, Hyderabad

Hyderabad

- Codified a scatter-gather integration across **25 hotel suppliers** on a **MuleSoft enterprise service bus**, succeeding where a prior **.NET 2.0** upgrade had failed
- Cut operational cost **46%** by retiring a third-party pricing product for an in-house **ELK** cache with a 20-minute time-to-live
- Reconciled supplier hotel identifiers onto **GIATA MultiCodes**, automating the pipeline on **AWS Lambda** with human approval retained, growing revenue **8% month over month**
- Lifted hotel booking conversions **65%** by widening matched inventory and ranking contracted properties first across **2–3 million daily requests**

Technical Architect

May '15 - Nov '15

CURA Software Solutions

Hyderabad

- Introduced **Linux/Unix deployment** for the Governance, Risk & Compliance platform, cutting running costs and winning **12 new customers within 2 months**
- Ported the data model from **Microsoft SQL Server** to **MySQL**, enabling the low-cost Linux deployment

Technical Lead

Jul '13 - May '15

innRoad India Hotel Software Pvt. Ltd.

Hyderabad

- Converted a **legacy ASP.NET 2.0** application into a modern **single-page application** using Knockout.js and a Boilerplate JavaScript framework
- Productized **REST-based WCF** services that enabled and monetised customer integrations
- Codified engineering standards and mentored the team through a full **product-modernisation effort**

Software Architect

Jun '12 - May '13

McDonald's Corporation

OakBrook, IL

- Modeled the schema and governance framework for an enterprise Master Data Management (MDM) program
- Forged interactive user-interface screens with drill-down, dynamic drop-downs, and full row-level editing
- Partnered with data owners to deliver a robust enterprise data-governance system across the organisation

Senior Java Developer

Apr '08 - May '12

El Paso Corporation

Houston, TX

- Rebuilt a legacy Citrix-hosted Java application into a modern **.NET 2.0** stack on **SQL Server 2005**

- Steered a scalable, maintainable .NET platform that replaced the outdated system while preserving core business functions
- Anchored large-scale enterprise modernisation for a high-stakes energy-sector client

.NET Developer

Apr '07 - Apr '08

LyntonWeb

Houston, TX

- Developed Paymaster, a **PayPal**-style payments application, for a Mexican financial institution

EDUCATION

PG Diploma in Artificial Intelligence and Machine Learning

Jan '21

IIIT Bangalore

Master's in Computer Science

Aug '04 - Dec '07

University of Houston

Houston, Texas

CERTIFICATION

PG Diploma in Artificial Intelligence and Machine Learning, IIIT Bangalore